

Retrieval-Augmented Generation with Multi-Agent Collaborative Search

Description

Large language models (LLMs) exhibit strong reasoning capabilities in solving human-level tasks. However, they are prone to hallucinations due to a lack of grounded knowledge. Retrieval-Augmented Generation (RAG) has emerged as a practical approach to mitigate this issue. Recent work has further integrated LLM reasoning into the RAG process, leading to iterative RAG methods [1, 2]. In these methods, the LLM serves as a search agent, dynamically interacting with the retrieval system to gather external knowledge and generate more reliable outputs. Despite their promise, these methods often rely on lengthy multi-round searches, which can be time-consuming and are vulnerable if any step in the reasoning chain fails to retrieve relevant supporting documents. In contrast, recent studies suggest that coordinated multi-agent collaboration can harness collective intelligence to surpass the performance of individual agents [3, 4]. This observation motivates the central question of this research: Can RAG systems benefit from multi-agent collaborative search? Specifically, multi-agent collaborative search allows parallel exploration of multiple search trajectories and facilitates the exchange of valuable findings through inter-agent communication, offering a more efficient and robust solution to RAG applications.

The goal of this project is to develop multi-agent collaborative search strategies that address the efficiency limitations of existing iterative RAG methods while improving the reliability of generated outputs. The potential research questions include, but are not limited to:

- Q1. Can multi-agent collaborative search increase the parallelism of iterative RAG, thereby improving both efficiency and overall performance?
- Q2. What collaboration strategies can maximise the utilisation of collective intelligence in multi-agent RAG systems?
- Q3. Can inference-time scaling laws be observed in multi-agent RAG systems?

The project falls under Machine Learning and Artificial Intelligence, highly related to Information Retrieval and Large Language Model Reasoning.

The project will be conducted at the Informatics Institute of the University of Amsterdam under the guidance of Yijia Zheng, a PhD student at the Multimedia Analytics Lab (<https://multix.io>).

This project is designed for students to complete their Master's thesis and does not pay for working hours. The student is expected to be comfortable programming in Python (especially LLM inference and training frameworks, like vLLM, Transformers, DeepSpeed, etc) and working with multiple kinds of datasets for RAG (e.g., multi-hop QA datasets and open-domain summarisation datasets).